

NEWTON SCHOOL

PRE-CALCULUS

ANSWER KEY

Inverse Trig Homework Answer Key: Real Numbers (Degrees & Radians)

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Standard: CCSS.MATH.CONTENT.HSF-BF.B.4 — Evaluate $\sin^{-1}$, $\cos^{-1}$, and $\tan^{-1}$ of standard unit-circle values. Express answers in both degree and radian measure.

Part A — $\sin^{-1}$ Answers

Problem 1

$$\sin^{-1}\left(\frac{1}{2}\right) = ?$$

$\sin^{-1}$

1Q value

$\sin 30^\circ = \frac{1}{2}$ (unit circle first quadrant). Since $30^\circ \in [-90^\circ, 90^\circ]$, it is in range.

$$30^\circ / \frac{\pi}{6}$$

Problem 2

$$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = ?$$

$\sin^{-1}$

1Q value

$\sin 60^\circ = \frac{\sqrt{3}}{2}$. Angle $60^\circ \in [-90^\circ, 90^\circ]$.

$$60^\circ / \frac{\pi}{3}$$

Problem 3

$$\sin^{-1}\left(-\frac{1}{2}\right) = ?$$

$\sin^{-1}$

negative value

$\sin(-30^\circ) = -\frac{1}{2}$. Negative input $\rightarrow$ negative output for $\sin^{-1}$. Angle $-30^\circ \in [-90^\circ, 90^\circ]$.

$$-30^\circ / -\frac{\pi}{6}$$

Problem 4

$$\sin^{-1}(0) = ?$$

$\sin^{-1}$

axis value

$\sin 0^\circ = 0$. The origin on the unit circle.

$$0^\circ / 0$$

Problem 5

$$\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = ?$$

$\sin^{-1}$

45° value

$\sin 45^\circ = \frac{\sqrt{2}}{2}$. Angle $45^\circ \in [-90^\circ, 90^\circ]$.

$$45^\circ / \frac{\pi}{4}$$

Problem 6

$$\sin^{-1}(-1) = ?$$

$\sin^{-1}$

boundary value

$\sin(-90^\circ) = -1$. This is the lower boundary of the $\sin^{-1}$ range.

$$-90^\circ / -\frac{\pi}{2}$$

Problem 7

$$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = ?$$

$\sin^{-1}$

negative value

$\sin(-60^\circ) = -\frac{\sqrt{3}}{2}$. Negative input $\rightarrow$ negative output. $-60^\circ \in [-90^\circ, 90^\circ]$.

$$-60^\circ / -\frac{\pi}{3}$$

Part B — $\cos^{-1}$ Answers

Problem 8

$$\cos^{-1}(0) = ?$$

$\cos^{-1}$

axis value

$\cos 90^\circ = 0$. Axis intersection at the top of the unit circle. $90^\circ \in [0^\circ, 180^\circ]$.

$$90^\circ / \frac{\pi}{2}$$

Problem 9

$$\cos^{-1}\left(\frac{1}{2}\right) = ?$$

$\cos^{-1}$

1Q value

$\cos 60^\circ = \frac{1}{2}$. Note: $\cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$, not 30° — remember cos and sin swap at 60° vs 30° .

$$60^\circ / \frac{\pi}{3}$$

Problem 10

$$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = ?$$

$\cos^{-1}$

1Q value

$\cos 30^\circ = \frac{\sqrt{3}}{2}$. Angle $30^\circ \in [0^\circ, 180^\circ]$.

$$30^\circ / \frac{\pi}{6}$$

Problem 11

$$\cos^{-1}\left(-\frac{1}{2}\right) = ?$$

$\cos^{-1}$

2Q value

$\cos 120^\circ = -\frac{1}{2}$. Negative cosine $\rightarrow$ 2nd quadrant angle within $[0^\circ, 180^\circ]$.

$$120^\circ / \frac{2\pi}{3}$$

Problem 12

$$\cos^{-1}\left(\frac{\sqrt{2}}{2}\right) = ?$$

$\cos^{-1}$

45° value

$$\cos 45^\circ = \frac{\sqrt{2}}{2}. \text{ Angle } 45^\circ \in [0^\circ, 180^\circ].$$

$$45^\circ / \frac{\pi}{4}$$

Problem 13

$$\cos^{-1}(-1) = ?$$

$\cos^{-1}$

boundary value

$$\cos 180^\circ = -1. \text{ This is the upper boundary of the } \cos^{-1} \text{ range.}$$

$$180^\circ / \pi$$

Problem 14

$$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = ?$$

$\cos^{-1}$

2Q value

$$\cos 150^\circ = -\frac{\sqrt{3}}{2}. \text{ Negative cosine } \rightarrow \text{ 2nd quadrant. Reference angle } = 30^\circ, \text{ so}$$
$$\text{answer} = 180^\circ - 30^\circ = 150^\circ.$$

$$150^\circ / \frac{5\pi}{6}$$

Part C — $\tan^{-1}$ Answers

Problem 15

$$\tan^{-1}(1) = ?$$

$\tan^{-1}$

1Q value

$\tan 45^\circ = 1$. Angle $45^\circ \in (-90^\circ, 90^\circ)$.

$$45^\circ / \frac{\pi}{4}$$

Problem 16

$$\tan^{-1}(\sqrt{3}) = ?$$

$\tan^{-1}$

1Q value

$\tan 60^\circ = \sqrt{3}$. Angle $60^\circ \in (-90^\circ, 90^\circ)$.

$$60^\circ / \frac{\pi}{3}$$

Problem 17

$$\tan^{-1}(0) = ?$$

$\tan^{-1}$

axis value

$\tan 0^\circ = 0$. Origin on the unit circle.

$$0^\circ / 0$$

Problem 18

$$\tan^{-1}(-1) = ?$$

$\tan^{-1}$

negative value

$\tan(-45^\circ) = -1$. Negative input $\rightarrow$ negative output for $\tan^{-1}$.
 $-45^\circ \in (-90^\circ, 90^\circ)$.

$$-45^\circ \quad / \quad -\frac{\pi}{4}$$

Problem 19

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = ?$$

$\tan^{-1}$

1Q value

$\tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$. Angle $30^\circ \in (-90^\circ, 90^\circ)$.

$$30^\circ \quad / \quad \frac{\pi}{6}$$

Problem 20

$$\tan^{-1}(-\sqrt{3}) = ?$$

$\tan^{-1}$

negative value

$\tan(-60^\circ) = -\sqrt{3}$. Negative input $\rightarrow$ negative output. $-60^\circ \in (-90^\circ, 90^\circ)$.

$$-60^\circ \quad / \quad -\frac{\pi}{3}$$